Excluding the interior two-bridge equality

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$, $n \ge 12$, be transitive with a connected string C-marking and all maximal omissions intransitive. Assume the high-rank equality configuration of the accepted nested-bridge theorem, with both crossings interior:

$$i = k-1, \quad h = k, \quad j = k+1, \quad r = k+l+1, \quad k, l \ge 2, \quad n = 2r.$$

The three regions are A, $M = \{b, c\}$, D, of sizes $2k, 2, 2l$. The unique crossing edges are $a -_i b$ and $c -_j d$. Write

$$L = G_{<i}, \quad s = \rho_i, \quad \tau = \rho_h, \quad t = \rho_j, \quad R = G_{>j}.$$

Here L fixes $M \cup D$ and R fixes $A \cup M$. We impose the two additional *literal* purity conditions

$$s|_D = 1, \qquad t|_A = 1. \tag{1}$$

These hold in the intended Proposition 5.28 routing when h is the last shared colour for the i-cut and $g = h$ the first shared colour for the j-cut: t already moves $M \cup D$ across its bridge and therefore cannot move A; likewise s already moves $A \cup M$ and cannot move D. We do not infer (1) from the numerical equalities alone.

The single middle edge is $b -_h c$. Thus

$$\tau|_M = (b\ c), \quad s = (a\ b)u_A, \quad t = (c\ d)v_D,$$

with u_A, v_D supported inside the indicated outside regions. The original omission orbits give

$$\langle L, \tau \rangle \text{ transitive on } A, \qquad \langle \tau, R \rangle \text{ transitive on } D. \tag{2}$$

All other generators fix those respective regions, by (1) and the stated prefix/suffix support conditions.

Lemma 1 (The exact critical side models)**.** *The involution τ exchanges k pairs in A and l pairs in D. On A, either L is transitive, or it has exactly two orbits of size k, each with faithful natural action S_k. On the k pairs of $\tau|_A$, its distinguished $k-1$ generators form an independent involutory string generating S_k. For odd $k \ge 5$ this is the ordinary adjacent-transposition path; for $k = 3$ it is the two-transposition path. The dual statements hold for R on D. If $k = 3$, the transitive alternative for L is impossible.*

Proof. The element τ centralizes L. If L is transitive on A, then its nontrivial centralizing involution is fixed-point-free. The pair action of L is transitive. Every distinguished generator is essential in that action: if a proper distinguished subset J had the same pair image, $\langle J, \tau \rangle$ would be transitive on A. A discarded generator is supported on A, so adjoining it to its global omission could not join omission orbits. That omission would already be transitive on Ω, a contradiction. Whiston's equality bound therefore makes the $k-1$ pair images an independent family generating S_k.

If L is intransitive, use the accepted faithful commuting-orbit normalization. Its orbit size is at least $k = i+1$, there are at least two orbits, and $|A| = 2k$. Hence there are exactly two k-point orbits. Whiston's equality theorem makes the faithful action on each S_k, and τ exchanges the

two orbits, identifying their actions. Again it is fixed-point-free and the pair action is the same natural S_k action.

In either case the independent involutory pair images form a string. For odd $k \geq 5$, the complete paid L047 maximal-string theorem and its degree-five certificate make them an ordinary path. For $k = 3$, the two involutions generating S_3 are distinct transpositions, so again give its ordinary path. The subgroup omitting the near endpoint fixes the pair $\{a, a^\tau\}$ pointwise, since it fixes a by the unique-crossing lemma and centralizes τ; this identifies the anchor endpoint of the path. No classification for even k is needed here, and no projected IP is assumed.

For $k = 3$ in the transitive alternative, identify the paired action with a subgroup of $C_2^3 \rtimes S_3$. Both distinguished generators are even on six points, so their binary flip vectors have even sum. Their product has pair image a 3-cycle. Its cube is a constant flip vector whose coordinate is the sum of the product's flip vector, and that sum is zero. Thus their product has order exactly three. Two involutions with product of order three generate S_3 of order six. A transitive action of that group on six points is regular, whereas its nonidentity far generator fixes a. This contradiction leaves precisely the two natural three-point orbits. Every argument reverses for the right side. □

Lemma 2 (No odd critical side)**.** *Under the standing hypotheses and (1), k is even. By reversal, l is even as well.*

Proof. First suppose $k \geq 5$ is odd. Let $H = G_{<i-1}$, the omission of the near endpoint of L. By Lemma 1, H fixes a and $a' = a^\tau$, and induces the full S_{k-1} on the remaining $k-1$ pairs of A. Those two anchors are its only fixed points in A. Since $s = (a\ b)u_A$ centralizes H and u_A preserves $A \setminus \{a\}$, it fixes a'. Its action on the other $2(k-1)$ points centralizes H. The complete accepted paired-row centralizer lemma, with even $k - 1 \geq 4$, says that this involution is even. Hence u_A is even and s is odd, contradicting $G \leq A_n$.

It remains to exclude $k = 3$. By Lemma 1, after identifying the two natural rows through τ, write

$$A = \{1, 2, 3, 4, 5, 6\}, \quad a = 3, \quad a' = 6,$$

$$x = \rho_0 = (1\ 2)(4\ 5), \quad y = \rho_1 = (2\ 3)(5\ 6), \quad \tau|_A = (1\ 4)(2\ 5)(3\ 6).$$

Here $i = 2$, so $s = (3\ b)u_A$ commutes with x. The point 6 is fixed by u_A: it is the sole x-fixed point in $A \setminus \{3\}$. On $\{1, 2, 4, 5\}$ an involution centralizing $(1\ 2)(4\ 5)$ either preserves both pairs or exchanges them. In the latter case it has two transpositions and is even. In the former case its odd possibilities are exactly one pair flip. Evenness of s therefore forces

$$u_A = (1\ 2) \quad \text{or} \quad (4\ 5). \tag{3}$$

Now omit x from the *global* group. The remaining τ, t, R are transitive on $M \cup D$: equation (2) makes $\langle \tau, R \rangle$ transitive on D, the edge $c -_t d$ joins c, and $b -_\tau c$ joins b. The s-edge $3b$ and the τ-edge 36 then add $3, 6$ to this omission orbit. The y-edges add $2, 5$. In the first case of (3) the other s-edge adds 1, and τ adds 4; in the second case it adds 4, and τ adds 1. Thus the omission is transitive on all Ω, contradicting our standing hypothesis. This excludes three, and every odd $k \geq 2$ has now been excluded. Reversing the entire argument proves the assertion for l. □

Theorem 3 (Interior touching bands are impossible)**.** *The equality $g = h$ in the nested-bridge high-rank theorem cannot occur with both unique crossings interior and with the literal purity conditions (1). In particular this equality is excluded when h and g are the last and first shared colours in the oriented Proposition 5.28 configuration.*

Proof. Lemma 1 and $M = \{b, c\}$ show that τ is a product of exactly $k + l + 1$ transpositions. Lemma 2 makes both k, l even. Thus τ is odd, the final contradiction. □

Attribution, dependencies and boundary. The source is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.28, physical page 37. The new proof is the critical side parity and literal omission argument above. It reuses the complete accepted nested-bridge count, commuting-orbit and pair image arguments, the whole L047 maximal-string theory and degree-five certificate, Whiston's bounds, and the paired-row centralizer lemma. No sample-only claim or new finite computation is a premise. The degree-six norm calculation is proved here directly.

Endpoint touching configurations, the separate $g > h$ middle-orbit equality and the remaining full-source routing and rigidity are not closed by this note. There is no new formal declaration, whole-original recovery or question/priority claim.